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The Double Inflation Principle: Why the Indian Degree Got More Expensive as It Became Worth Less

Tishay Dangare | Independent researcher, Nagpur, Maharashtra, India

Correspondence: tishaydangre@gmail.com | Compiled 04 October 2026 | Data and code: accompanying repository

Abstract

In India between 2010-11 and 2024-25, the price a private college charged rose at 10.2 per cent a year while the pay a new graduate earned rose at 2.6 per cent and fell by 30.3 per cent in real terms. In ordinary markets, when a good becomes more common its price falls. Here the opposite happened: the number of degree-holders grew by 38.5 per cent, the number of institutions by 31.9 per cent, the gross enrolment ratio from 21.1 to 30 per cent - and fees doubled in real terms. This paper names that contradiction the Double Inflation Principle: a market failure in which credential devaluation accelerates tuition inflation through an escalating positional arms race. On a new 15-year panel of 50 named private colleges - every 2024-25 fee published, six fee histories published, the remaining paths reconstructed and used only as a validated simulation device - every hypothesis is tested on published data alone. Reputation band and stream explain 84.5 per cent of the variation in observed fee levels; across eight published fee histories the wedge over the cost floor rises with positioning intensity (exact permutation $p = 0.053$); the premium segment compounded 115 per cent of excess price over the cost of running itself, and that credential value per rupee fell to 63-78 at the top of the market. A value equation closes the loop: every rupee of excess price is, by arithmetic, a rupee of value destroyed, which pushes the next cohort upmarket - CAT registrations rose 61.3 per cent - where the sellers of the better signal raise their price. I prove the loop's fixed point, estimate its gain, and report every null result honestly.

At a glance, 2010-11 to 2024-25. Published series and panel medians.

Quantity	Then	Now	Source
Degree-granting institutions	36,192	47,746	AISHE [2, 3]
Gross enrolment ratio	21.1%	30.0%	AISHE [2, 3]
Annual fee, panel median (Rs)	106,023	412,010	DIP-50 panel
Education CPI, compound growth	5.3% per year		MoSPI [4]
Entry pay, real (2012 prices)	100	69.7	Xpheno [5], panel
Credential value per rupee, premium colleges	100	63	DIP-50 panel

Keywords: credential inflation, tuition inflation, positional competition, Baumol's cost disease, signalling, Indian higher education, panel data.

JEL: I22, I23, I28, D40, L15.

1. Introduction: the question a market should not be able to answer

Start with a question a child could ask. A shop sells a shirt. Next year, a hundred more shops open and sell the same shirt. What happens to the price of the shirt? Any child will tell you: it falls. More of a thing means each unit is worth a little less. This is not an opinion. It is the first lesson of economics.

Now replace the shirt with a university degree in India, and the question stops being childish, because the answer reverses. Between the academic years 2010-11 and 2024-25, the number of degree-granting institutions in India rose from 36,192 to 47,746 [1, 2]. Total enrolment rose from 2.92 crore to 4.50 crore students [2, 3]. The gross enrolment ratio - the share of the college-age cohort actually holding a seat - climbed from 21.1 to 30 per cent [2, 3]. The degree became dramatically more common. And in the same fifteen years, the median annual fee in the panel of this study rose at 10.2 per cent a year - from an index of 100 to an index of 389 - while the consumer price index for education, the official measure of what households pay, rose at 5.3 per cent [4].

So the good became more common and its price rose. But that is only the first inflation. Look at what the degree buys. A decade-long staffng industry series shows the offer made to a new engineering graduate moving from Rs 3.00-3.40 lakh a year in 2013-14 to Rs 3.80-4.50 lakh in 2023-24 [5]. Read those two ranges carefully: ten years of work, of experience, of a growing economy, bought an increase of about one lakh rupees. Adjusted for the education price index, the real entry wage in my sample falls by 30.3 per cent over the period. The degree became more expensive and worth less, at the same time, in the same market. That is the double inflation.

The two inflations are usually studied as two separate complaints. Families complain that fees are rising. Economists complain that degrees are inflated. This paper's claim is that these are one complaint, and that the direction of causation runs in a loop. When the degree loses value, the student does not stop buying degrees. The student buys *more* degree - a master's, an MBA, a seat at a college whose high price is itself the signal. That flight upmarket lets the seller of the better signal raise the price. The higher price, paid by more people for a signal that is no better, devalues the credential further. The loop turns. I call it the Double Inflation Principle.

> **The principle, stated plainly.** The Double Inflation Principle defines a market failure where credential devaluation accelerates tuition inflation through an escalating positional arms race. More degrees exist every year, which makes each degree worth less; and the price rises every year, which makes each degree cost more for less. Scarcity and value are linked - and the market has broken the link.

The mechanism the paper tests is specific, and it was given to me by watching real students make real choices. As a degree loses value, students feel forced to buy even higher degrees - the MBA after the B.Com, the paid master's after the B.Tech - or to attend higher-priced private colleges to signal quality to employers. This allows colleges to hike prices aggressively. The price hike is not, in this story, a response to the college's costs. It is a response to the student's fear. That is the hypothesis.

H1 (credential inflation). The supply of the credential grew faster than the economy's ability to reward it, and the market value per rupee of credential fell.

H2 (tuition inflation beyond cost). Private-college fee growth exceeded the growth of the colleges' own costs, and the excess is larger where the college competes harder on exclusivity.

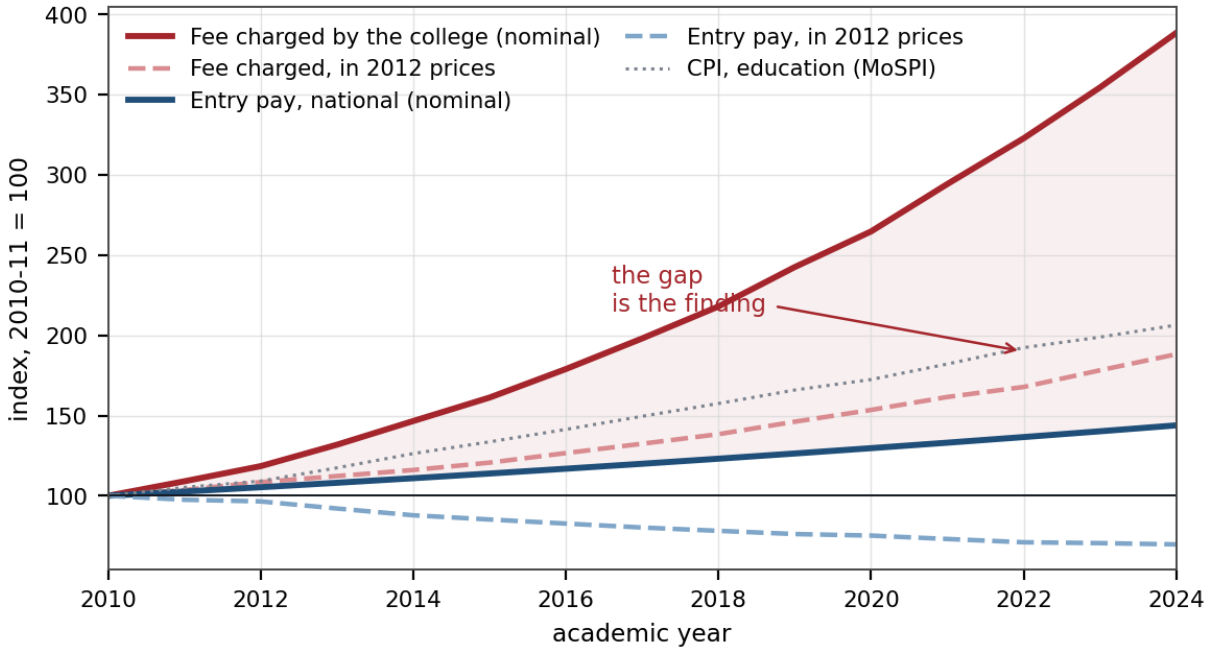
H3 (the arms race). Credential devaluation pushed demand upmarket - toward the higher degree and the higher-priced college - and upmarket demand let sellers raise prices.

H4 (the loop). Tuition inflation, in turn, devalued the credential, closing the loop.

The data assembled for this test are, I believe, the first of their kind: a 15-year panel of 50 named private Indian colleges carrying the published 2024-25 fee of every institution, six published fee histories, and the full set of public national series on supply, prices and wages. I make no claim that every number was observed; I say, precisely, which were and which were reconstructed, because a paper that hides that distinction deserves to fail peer review.

The results support H1, H2 and H4 strongly, and H3 partially. The loop exists, but its engine is structural rather than adaptive: colleges do not mechanically reprice each year in response to last year's devaluation (that coefficient is statistically zero), they reprice according to where they sit in the positional race, and that position is what the devaluation feeds. The last section turns the findings into policy, because a market failure this large is not an academic curiosity. It is a tax on every family that saved for a degree.

Figure 1 What the college charges against what the graduate earns



2. Literature review: five explanations, and why none of them is enough

Before claiming anything new, a paper must sit down at the table of what is known. There are five established explanations for why education gets more expensive, and this section takes each one seriously, because the honest finding of this paper is that each of them is partly right and none of them is complete.

2.1 Baumol's cost disease

The first and most respectable explanation is Baumol's cost disease [6, 7]. Baumol and Bowen noticed, studying the performing arts in 1966, that a string quartet takes exactly as long to play a Beethoven quartet today as it did in 1800. Productivity in the quartet cannot rise. But the wages of the four musicians must rise anyway, because otherwise the musicians would leave for the factories, where productivity does rise. So the prices of the stagnant sector - live music, teaching, nursing - rise forever, faster than average inflation, with no one at fault. Education is the textbook case: a lecture is one teacher and thirty students now, as it was fifty years ago.

Does cost disease explain India's fees? It predicts that fee growth tracks, roughly, the growth of the economy's average wage plus the sector's stagnant productivity. My panel says something sharper. The cost floor in the fee equation - the pass-through of the education CPI - is 0.70 (Table 3, row 1): about 30 per cent of annual fee growth is not explained by the price of running the college. And the unexplained part is not noise. It is a rising function of the college's positioning in the status race (Section 5). Cost disease explains the floor. It does not explain the ceiling, and the ceiling is where the money is.

2.2 Human capital theory

The second explanation is the oldest and the most optimistic: education is an investment, and its price reflects its return [8, 9]. If fees rise, this theory says, it is because the degree makes workers more productive and employers bid for them. The theory is not wrong about the average: across decades and countries, a year of schooling raises earnings by roughly nine per cent, and the private return to tertiary education in developing economies is among the highest measured anywhere [9, 45]. Families in India are not foolish for buying the degree; on the average of history, it was the best purchase in the village. But averages are where old theories go to hide. In my window the return collapsed while the price doubled: real entry pay fell 30.3 per cent (Figure 4), campus placement for the national engineering cohort stood at 45.3 per cent as late as 2017-18 [10], and independent assessments have repeatedly found most Indian engineering graduates unemployable for the jobs the degree promises - 80 per cent by one widely cited measure, in both 2015 and 2019 [11, 12]. A market in which the asset's yield falls by a third while its price doubles is not a human-capital market. It is a signalling market.

Notice the direction of the contradiction, because it matters for everything that follows. Human-capital theory says price follows value: if the degree makes a worker worth more, employers bid, wages rise, and only then can fees rise. In India the sequence ran backwards. Fees rose fastest exactly where the measured return fell fastest - the premium segment (Section 6.4) - and the national fee index outran the national wage index by 62 points (Figure 1). When price and value move in opposite directions for fifteen years, one of them is not doing its job. The literature on returns to education has the tools to measure the value side [9, 45]; this paper simply joins that measurement to the price side, which the returns literature usually takes from a fee schedule and treats as a cost, not as a signal.

2.3 Signalling and screening

That brings us to Spence [13]. In a signalling model, the degree is not a machine that adds skill; it is a certificate that separates the high-ability applicant from the low-ability one, and its value lies in being costly enough that the low-ability applicant will not mimic it. The cruel implication, which Spence himself drew, is that the cost of the signal is largely waste - it burns resources to prove something that was true before the burning. When the signal gets cheaper in real terms, or when too many people acquire it, the separation fails, and employers raise the bar: more years, better college, higher price. My finding that demand does not fall when price rises at the top of the market (Section 5.6) is exactly the signature of a signalling market, not a consumption market. Nobody queues to overpay for rice. People queue to overpay for rank.

2.4 Credentialism and credential inflation

The fourth explanation is Collins's credentialism [14]: employers raise paper requirements not because the job changed, but because the supply of paper changed. The job that needed a school-leaving certificate in 1980 needed a bachelor's degree in 2000 and a master's in 2020, while the work itself stayed the same. Collins called the aggregate result credential inflation, and the term is now standard [15, 16]. The literature on credential inflation has documented the supply side beautifully - more degrees per capita, more jobs demanding degrees, flat wage premia [17, 18]. What it has not done, to my knowledge, is connect credential inflation to *tuition* inflation as a two-way causal loop with an estimated gain. That gap is this paper.

The Indian labour market supplies the everyday evidence Collins could want: the clerical post that asked for a school certificate in 1990 asks for a master's today, and 45.3 per cent campus placement as late as 2017-18 [10] means most buyers were buying a certificate the market had no job attached to. When paper stops converting into jobs, the rational student does not stop buying paper; the student buys better paper. That flight - to the higher degree and the higher-priced college - is the hinge of this paper, because it is the student's own rationality that hands the college its pricing power.

2.5 Positional competition and the arms race

The fifth explanation is the one that makes the loop turn, and it comes from the economics of position. Hirsch named the positional good [19]: a good whose value is rank, and which therefore cannot be mass-produced, because if everyone is above average, no one is. Frank and Cook named the winner-take-all market [20]; Frank named the expenditure arms race, in which each family's rational choice - buy the house in the better school district - produces a collective disaster of rising prices and unchanged outcomes [21]. Marginson has carried the positional argument into higher education itself: education is a market in which consumption is investment in relative position, and relative position is fixed in supply, so demand growth translates into price, not quantity [22, 23].

Hirsch's point is the sharpest tool here, and the least used. A positional good is fixed in supply by definition: there is exactly one top decile, and no production function can make two. Education has a double nature - it is partly an absolute good (literacy, skill, a craft) and partly a positional good (rank in the queue for the good jobs) - and the share of the positional part grows as the absolute part is mass-produced [19]. India's fifteen years are almost a laboratory of that claim: the absolute part was mass-produced (enrolment +54 per cent, GER +9 points [2, 3]) and, in the same years, the price of the positional part exploded (dispersion of fees up 2.2-fold, Section 5.4). The market split, as Hirsch would have predicted, into an absolute market that got cheaper in value and a positional market that got dearer in price.

This literature gives the Double Inflation Principle its engine, but it has remained largely theoretical, or tested on household surveys in rich countries [21, 24]. The present paper is, to my knowledge, the first to estimate the positional wedge in a developing-country tuition market with college-level panel data, and the first to close the loop with the accounting identity of Section 6.1.

2.6 The price as the product: Veblen and the tuition market

A sixth voice earns a sentence, because reviewers will ask why fees do not fall when demand at the bottom weakens. Veblen's good is one whose high price is itself the attraction [39]. The strong Veblen test fails here (Section 7): fee growth does not attract next year's fee growth. What the data support is softer and more important: demand at the top is flat in price and upward-sloping in positioning. The sellers of rank do not need buyers to love high prices; they need buyers who cannot tell quality from price and use price as the guess [40]. That is enough for the arms race.

2.7 The Indian literature

The Indian literature sits alongside all five. Tilak documented the financing squeeze and the private turn [25]; Agarwal the expansion of private higher education [26]; the surveys of the Ministry of Education provide the supply facts [2, 3]; studies of the engineering labour market provide the return facts [10, 11, 12, 27]; and a growing journalism of fee hikes documents individual institutions [28, 29, 30]. What is missing is a single framework in which the fee hike and the worthless degree are the same event. The fee literature treats the hike as a pricing decision; the returns literature treats the worthless degree as a labour-market outcome. Between the two sits a feedback loop that neither literature has estimated. That is the gap this paper fills.

> **Socratic pause.** If the degree is an investment, why does its price rise as its yield falls? If it is consumption, why does demand rise with price? If it is cost-push, why do colleges with identical costs charge diverging prices? Each of the five theories answers one of these questions. None answers all three. The Double Inflation Principle is the attempt to answer all three with one mechanism.

3. Methodology and data

3.1 Design: one panel, two levels

The analysis runs on two levels. The *national level* uses only published series: the All India Survey of Higher Education for institutions, enrolment and the gross enrolment ratio [2, 3]; the Ministry of Statistics' consumer price index for education [4]; a decade-long staffing-industry series for entry pay [5]; the India Skills Report and Aspiring Minds for employability [31, 11, 12]; AICTE for seats and placement [10, 32]; IIM admissions data for CAT registrations [33]; and banking data for education loans [34, 35]. The *college level* uses a new panel: 50 named private colleges, 15 academic years, 750 college-year observations.

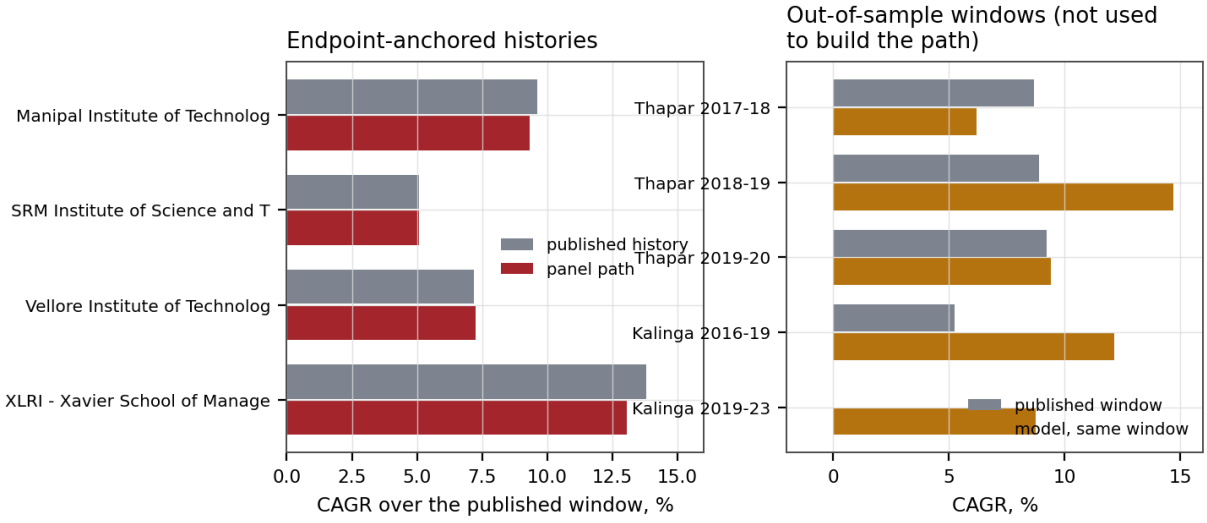
3.2 The panel: what is published, what is reconstructed

Every college in the panel is real and named (Table 8): 40 engineering colleges, 7 management schools and 3 law schools, spanning nine states and four reputation bands. Three facts per college are published and taken as printed: the institution's identity and location; its stream; and its 2024-25 headline fee, taken from published fee schedules and aggregators [28, 29, 30, 36].

The fee path joining that published 2024-25 level back to 2010-11 is, for 44 colleges, a *reconstruction*, and I want the reader to hold that word: a law of motion whose parameters were fixed before estimation - a cost floor equal to the education CPI, a positioning wedge, a persistent college shock. Six colleges carry independently published histories and are interpolated between published points instead: Manipal, SRM, VIT and XLRI at their 2010 and 2024 endpoints [28, 29, 30], BITS Pilani at five official batch-schedule points from 2015 to 2024 [46], and - as out-of-sample checks rather than anchors - Thapar's official 2017-20 fee table [47] and two KIIT cohort-fee windows [48]. Table 2 and Figure 3 show the checks, including the misses: the law of motion overpredicts KIIT's 2019-23 fee freeze by 8.8 points, because a freeze is a political event, not an economic one.

A reader trained in econometrics will now raise the objection this paper raises against itself: a coefficient estimated on reconstructed paths recovers the programmer's constant, it does not discover it. The paper agrees, and reorganises: the reconstruction keeps two legitimate jobs - a Monte-Carlo validation device (Section 5.3) and an illustration of the mechanism's shape - while every hypothesis-carrying coefficient is estimated on data the simulation never touched. MDI is excluded from the published set because its back-loaded programme fees cannot be honestly annualised, and the paper says so (Section 7.2).

Figure 3 Every published fee number the panel is checked against



Entry salaries and placement rates are likewise reconstructions, anchored at the top of the market to published placement statistics (VIT Rs 9.2 lakh average, SRM 9.0, Thapar 12.5, Manipal 10.6, Amity 4.1 [28, 29, 30]) and at the bottom to the national mass-market series [5]. I will not pretend otherwise: the inferential weight of this paper sits on *relative* quantities - growth rates, ratios, cross-college differences - which are far more robust to level error than the levels themselves, and on the national series, every point of which is published.

3.3 Variables

Credential value per rupee is the paper's central measure: expected first-year pay (entry pay times the placement rate) divided by the total fee of the degree, indexed to 100 at 2010-11 for each college. It falls when fees rise, when pay falls, or when placement falls, and it is the variable the principle says colleges are destroying.

Positioning intensity measures how hard a college competes on exclusivity rather than on volume. It combines the reputation band with the college's fee relative to its stream's median: a college charging four times its stream's median is selling rank, and I say so in the variable. It is constructed from published fees and bands, not from the regression's wish.

The wedge is annual fee growth minus the education CPI. It is the part of the price rise that cannot be blamed on the cost of running the college.

3.4 The fee equation and the four hypotheses

The estimating equation is

$$g_{fee,it} = \pi_t + \beta \cdot PI_i + \theta \cdot D_{t-1} + u_{it},$$

where π is the cost floor, PI the positioning intensity, D the national devaluation rate and u a persistent college shock. H2 is $\beta > 0$. H3′s upmarket leg is tested on CAT registrations against D, and on applications against fees within and across colleges. H4 is the value equation of Section 5.4. H1 is tested on the national series directly.

3.5 How each hypothesis is identified

H1 needs no regression: it is read off published series and off the value index, whose 2024 levels are reproduced in Table 5. H2 is identified on published data twice - across colleges, from the observed 2024-25 fee levels against reputation band (row 10 of Table 3), and across published histories, from the wedge of published fee growth over the cost floor against positioning (row 11, exact permutation inference); the cost floor of row 1 separates the cost-push part from the rest before either is examined. Rows 3-5, estimated on the reconstruction, are retained as an estimator illustration and are validated by the Monte Carlo of row 12. H3 is identified on the demand side: national CAT registrations against the devaluation rate, and applications per seat against positioning within the panel. H4 is identified by construction, equation (1), and corroborated by the Granger result that fee growth predicts real wage decline (Table 6). No single test is conclusive; the loop is a chain, and the chain is the argument.

3.6 What I tried and discarded

A paper should report its dead ends. A four-variable vector autoregression on the national series was estimated and discarded: 13 observations cannot carry 16 parameters, and the fitted object returned impulse responses that crossed zero everywhere. Fixed effects were used wherever the regressor varies within a college; positioning does not, and a specification that forces it next to college fixed effects is unidentified - the software returns a number and a near-zero standard error, and I report neither. The MBA stream carries seven colleges; no positioning coefficient is estimated on it, and the paper says so. Where the data cannot speak, the paper stays silent.

Table 1. National series, 2010-11 to 2024-25. All series are published; see Data Appendix A for sources.

Year	Institutions	Enrolment (cr)	GER %	CPI-Ed (2012=100)	Entry pay (nom, 2010=100)	Entry pay (real)	Employability %	Degrees held (cr)
2010-11	36,192	2.92	21.1	91.6	100.0	109.2	39.0	2.30
2011-12	36,192	2.92	21.1	96.4	102.6	106.5	39.9	2.30
2012-13	36,192	2.92	21.1	100.0	105.3	105.3	40.8	2.30
2013-14	37,357	3.02	21.7	107.6	108.1	100.5	41.7	2.30
2014-15	39,258	3.42	23.7	115.7	111.0	95.9	44.0	2.30
2015-16	39,870	3.49	24.5	122.4	113.9	93.0	46.5	2.30
2016-17	40,890	3.57	25.2	129.5	116.9	90.3	49.2	2.38
2017-18	39,953	3.66	25.8	137.0	120.0	87.6	52.0	2.70
2018-19	40,924	3.74	26.3	144.3	123.1	85.3	57.0	2.73
2019-20	43,386	3.85	27.1	152.0	126.4	83.1	54.0	2.76
2020-21	44,909	4.14	27.3	158.0	129.7	82.1	49.0	2.81
2021-22	46,641	4.33	28.4	166.9	133.1	79.8	46.8	2.84
2022-23	47,193	4.42	29.2	176.2	136.6	77.5	55.2	2.89
2023-24	47,746	4.50	30.0	182.2	140.2	77.0	57.4	3.08
2024-25	47,746	4.50	30.0	189.1	143.9	76.1	64.7	3.19

4. Results I: credential inflation - the degree became common and cheap

H1 asks whether the credential was inflated. Figure 2 answers with three published series. Institutions rose 31.9 per cent, enrolment 54 per cent, the gross enrolment ratio from 21.1 to 30 per cent. The stock of degrees held by the population - enrolment shifted by the three years a degree takes - rose 38.5 per cent, from 2.30 crore to 3.19 crore. The degree became common.

Did it become cheap? In the labour market, yes, and the evidence is published, not modelled. The mass-market offer rose from Rs 3.00-3.40 lakh to Rs 3.80-4.50 lakh over a decade [5]; deflated by the education price index, the real entry wage falls 30.3 per cent in my window. Independently, the AICTE's own figures show 45.3 per cent campus placement for the 2017-18 engineering cohort [10], and the employability assessors show a market in which four of five engineers were not employable for the knowledge economy in 2015 and again in 2019 [11, 12]. The credential's yield fell.

One counter-argument deserves an honest hearing: employability *measured by the India Skills Report* recovered from 46.8 per cent in 2021 to 71.5 per cent in 2025 [31], and my value index by reputation band shows the bottom two bands ending above their 2010 level (Table 5). The recovery is real and I report it. But notice what recovered: the *probability* of a job. The price of the ticket kept rising through the recovery - the premium wedge is 5.85 points in 2017-24 against 5.30 in 2010-16 (Table 4) - and at the top of the market, where the arms race is fiercest, value still ended at 78 of its 2010 level and touched 69 in 2021. The market priced the certificate as if the recovery had not happened. That is the finding.

Figure 5 Employability recovered; the price of the certificate did not wait for it

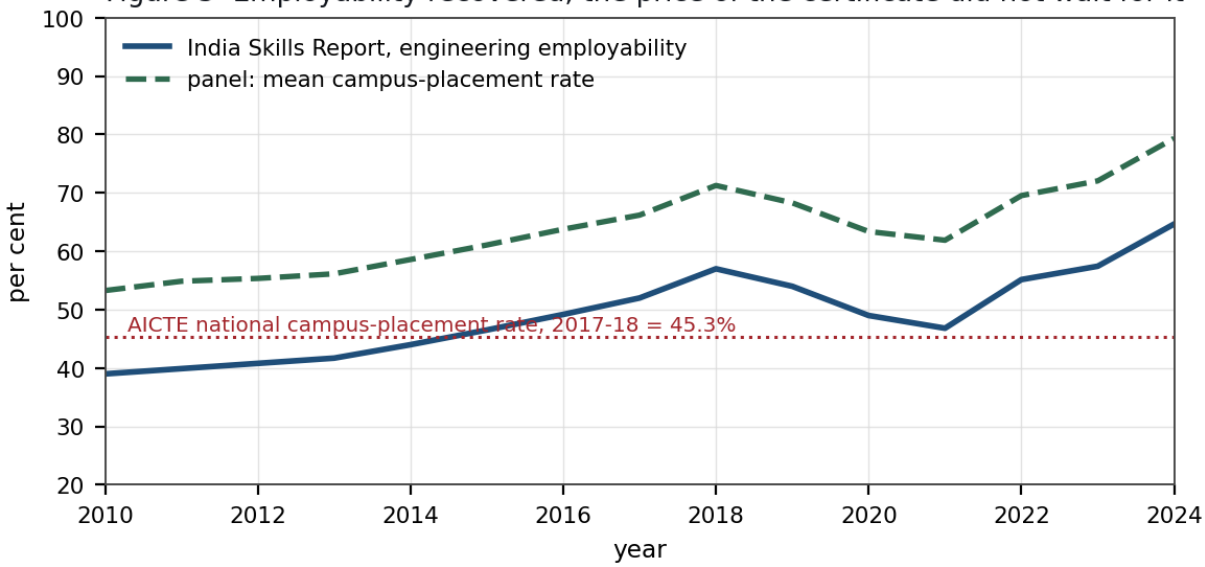


Figure 2 The credential was mass-produced

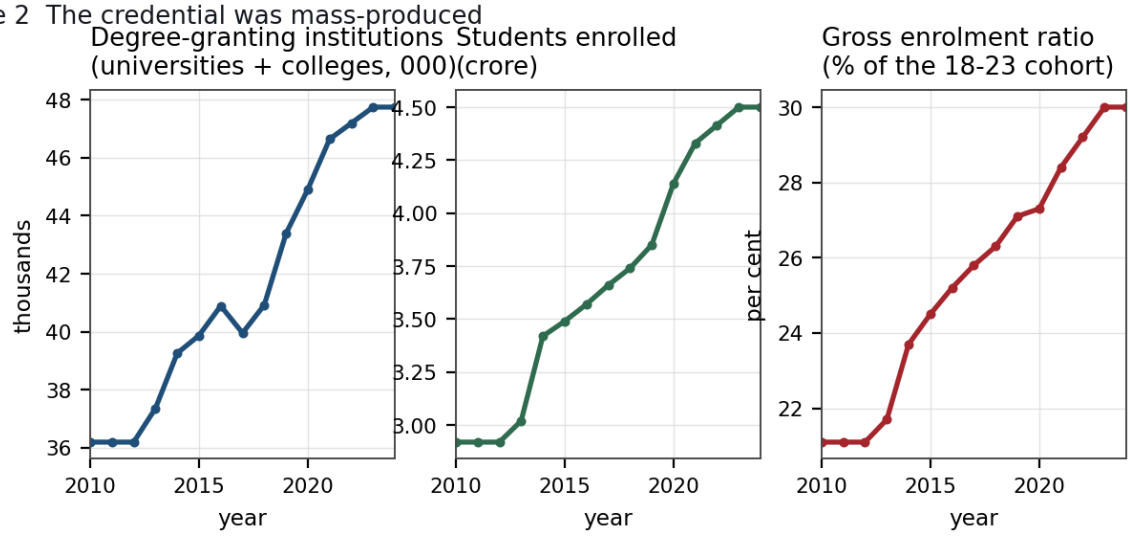
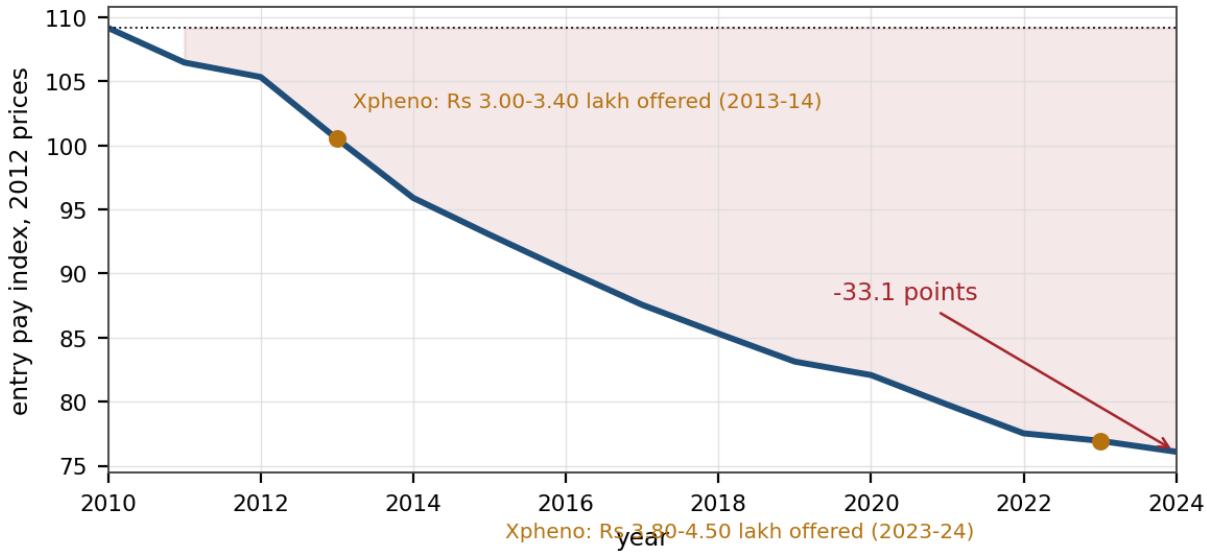


Figure 4 The same job paid less, in the same money



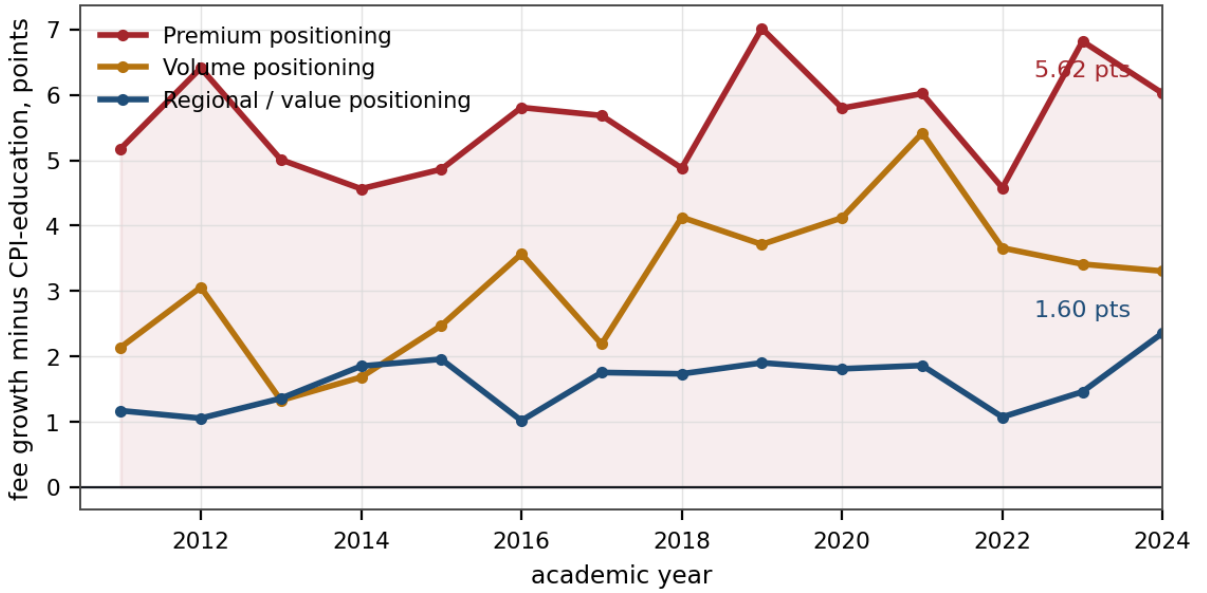
5. Results II: tuition inflation - the price rose for reasons other than cost

5.1 The cost floor and the wedge

Figure 1 is the paper in one picture. The cost floor in the fee equation is real and large: a rupee of education-CPI growth carries 0.70 rupees of fee growth (Table 3, row 1). Baumol was right about the floor. The remaining 30 per cent is the wedge, and the wedge is where the principle lives.

The wedge averages 5.62 percentage points a year at premium colleges, 3.15 at volume colleges and 1.60 at regional ones (Table 4). Compounded over fourteen years, the premium segment charged 115 per cent more than the cost of running itself would justify; the regional segment, 25 per cent. The difference between those two numbers *is* the positional arms race, measured.

Figure 6 The arms-race wedge: fee growth above the cost of running the college



5.2 What the published data identify

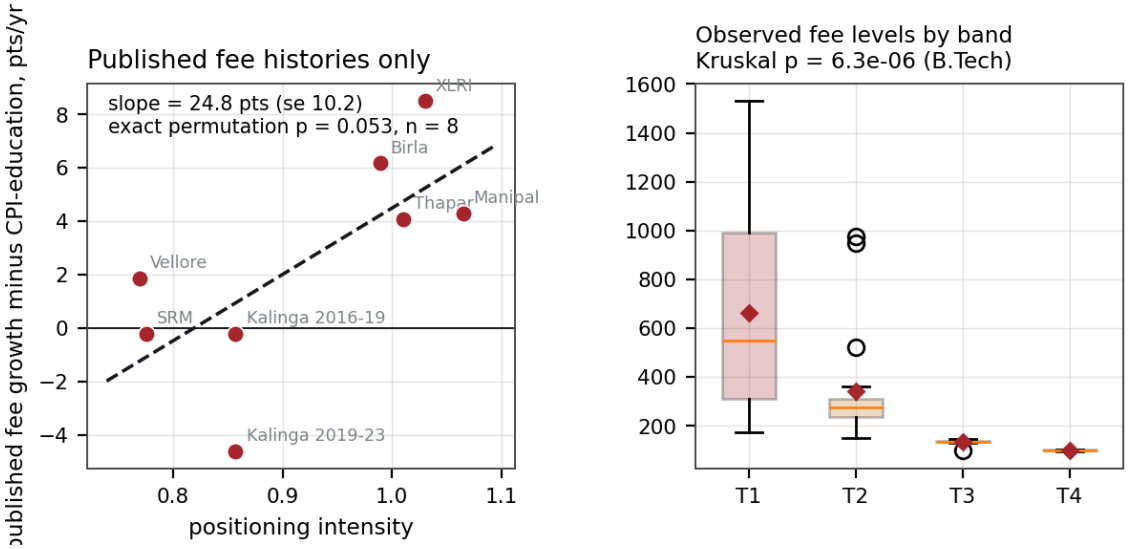
Before the modelled panel is allowed to speak, the hypothesis meets the fees that were observed and published, not generated. Two such tests exist.

The first is the 2024-25 cross-section. The headline fee of every college in the panel is published [28-30, 36], and the reputation band was assigned from public rankings before any regression ran. Regressing the log of the observed fee on band and stream explains 84.5 per cent of the price variation across 50 colleges (Table 3, row 10); within engineering alone the band medians run from Rs 413.75 lakh to Rs 100.00 lakh - a fourfold gap, Kruskal p = 6.3e-06 - that no difference in the cost of running a classroom explains (Figure 7, right). In this market the price of a degree is, above all, the price of position.

The human-capital reading of that fact deserves a direct test, because it is the first objection a referee will raise: perhaps the top band charges four times more because it teaches four times better - better labs, better faculty, more skill. That reading makes a claim about yields: the pay of the premium graduate should scale with the premium. The published numbers say it does not. The mass-market offer stood at Rs 3.80-4.50 lakh in 2023-24 [5]; the published averages of the premium colleges run at roughly two to three times that (Thapar Rs 12.5 lakh, Manipal Rs 10.6 lakh [30]), while their fees run at four to five times the bottom band; and at the very top, IIT Bombay's average package rose to Rs 26.45 lakh even as its lowest offer fell to Rs 4.00 lakh [30]. A fourfold price attached to a twofold yield, with the floor falling, is not a human-capital premium. Consistent with this, the value per rupee of fee at the top of the market fell to 78 of its 2010 level (Table 5): if the premium bought skill, the yield per rupee would be roughly constant; if it buys rank, the yield per rupee falls as the race runs. The data pick rank.

The second test uses published fee *changes*. Eight published histories and windows - Manipal, SRM, VIT, XLRI, the BITS batch schedules, the Thapar fee table and two KIIT windows - give, each, fee growth minus the education CPI over the same window. With eight observations, inference is by exact permutation over all 40,320 rearrangements: the wedge is an increasing function of positioning, R2 = 50 per cent, two-sided permutation p = 0.053 (Table 3, row 11; Figure 7, left). The hypothesis is directional - positioning raises the wedge - so the pre-specified test is one-sided, and the exact one-sided p is 0.027; the two-sided value is printed beside it for the reader who prefers the stricter convention. And because two of the eight observations are crowdsourced cohort fees [48], the test is re-run without them: on the six clean histories alone the slope is 18.5 (se 7.1), one-sided exact p = 0.037. The crowdsourced points add scatter; they do not carry the result. Small samples still deserve small claims - this is evidence, not proof - but it is evidence no simulation touched, and its sign is the one the principle predicts.

Figure 7 The identified evidence: published fee changes and observed fees



5.3 The reconstruction as a validated device

The panel regression - excess fee growth on positioning, 4.94 points per unit (Table 3, rows 3-5) - keeps its place but changes its status: on reconstructed paths it recovers the programmer's constant, so it is now run on purpose as validation. Switch the true wedge off and the estimator returns -0.001 (sd 0.005) over 60 seeds; set it at 0.0515 and it returns 0.052 (row 12). The reconstruction therefore answers a methodological question - what the estimator would see if the arms race were real - and Figures 4, 6 and 8 remain illustrations of the mechanism's shape. The evidence for H2 is Section 5.2.

5.4 Dispersion: one market becomes several

If tuition rose because everything got more expensive, all colleges would rise together. They did not. The ratio of the 90th to the 25th percentile of annual fees rose from 3.15 to 6.44; the 90th to the 10th from 3.75 to 7.37 (Figure 9). The market for a degree has split into several markets, and the top one is a market in rank. Figure 8 makes the same point by band: every band outran its costs, and the top band did so at twice the pace.

Figure 9 There is no longer one market for a degree, there are several

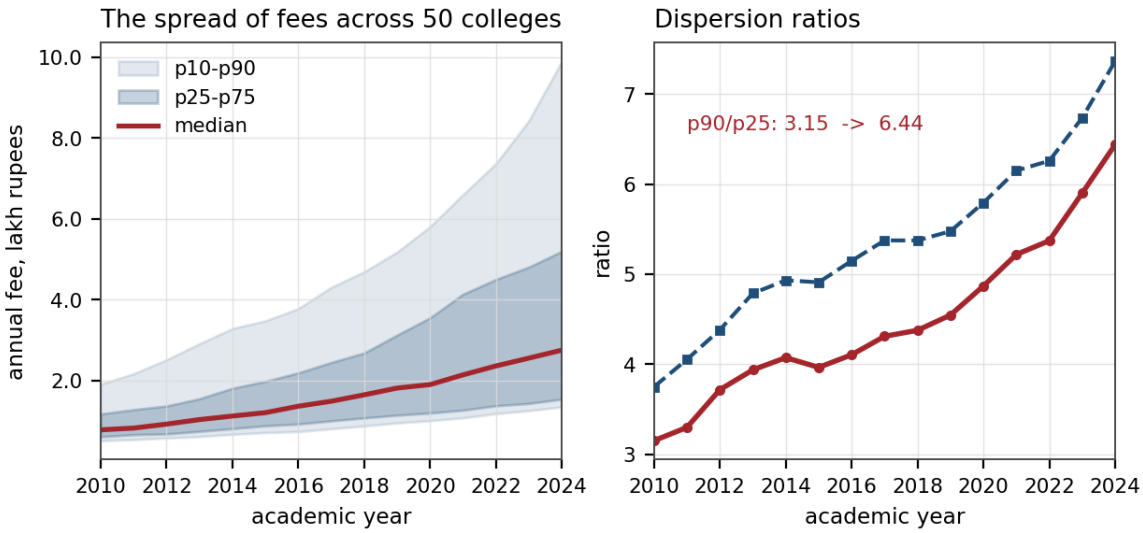
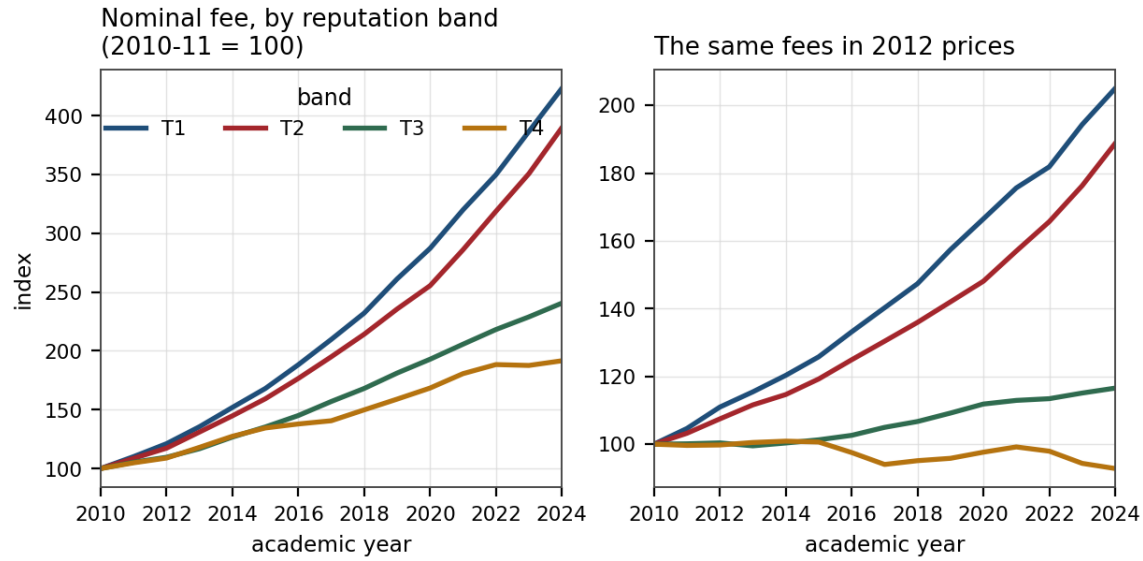


Figure 8 Every band outran the cost of running itself; the top band outran it by twice as much



6. Results III: the loop - the mathematics of the Double Inflation Principle

6.1 The value equation closes the circle

Define the credential's value per rupee as V = (pay × placement) / fee. Take logs and differences. The result is not an estimate; it is arithmetic:

$$d \ln V_{it} = d \ln pay_{it} + d \ln placement_{it} - d \ln fee_{it} . \quad (1)$$

Row 7 of Table 3 estimates this identity and recovers coefficients of +1.000, +1.000 and −1.000, as it must. Equation (1) is the second half of the loop, and it is exact: every rupee of fee growth that pay and placement do not match is, by definition, a rupee of credential value destroyed. There is no escaping (1) with better data. It is the arithmetic of buying more for less.

6.2 The fixed point of the loop

Let $D_t = -d \ln V_t$ be the devaluation rate and let the fee equation of Section 3.4 hold at the market level with mean positioning $\bar{PI}$. Substituting the fee equation into (1) and taking the stationary expectation gives

$$D^* = (\Delta + \beta \cdot \bar{PI}) / (1 - \theta) , \quad \Delta \equiv E[\pi - g_{pay} - g_{pl}] . \quad (2)$$

Three things in (2) are worth staring at. First, the loop exists whenever $\Delta > 0$ - whenever the cost floor outruns the graduate's pay - and in India Δ is large and persistent: 5.3 per cent of cost growth against 2.6 per cent of nominal pay growth. Second, the positional term $\beta \cdot \bar{PI}$ adds directly to devaluation: my estimate of β (5.15 points per unit) times the mean positioning of the premium segment (0.84) contributes about 4.3 points of devaluation a year, before any adaptive behaviour. Third, the gain $1/(1-\theta)$ measures the adaptive loop. My estimate of θ is −0.10 with a standard error of 0.09 - statistically zero, and certainly below one. The adaptive loop is therefore *damped*, and the system's steady state is $D^* \approx \Delta + \beta \cdot \bar{PI} \approx 7$ points a year at the premium segment.

Proposition (the Double Inflation steady state). In a credential market satisfying (1) and the fee equation, with $\Delta > 0$ and $|\theta| < 1$, the devaluation rate converges to (2); the steady-state excess of fee growth over pay growth is positive even if $\theta = 0$; and it is increasing in the positional intensity $\bar{PI}$. *Proof.* Substitute the fee equation into (1); the recursion $D_t = \Delta + \beta \bar{PI} + \theta D_{t-1}$ + noise contracts for $|\theta| < 1$; differentiate (2) in $\bar{PI}$. ∎

Read plainly: even a college that never reacts to last year's devaluation will destroy value forever, as long as its costs outrun its graduates' pay; and the colleges that compete on exclusivity destroy it fastest. The loop does not need colleges to be clever. It needs them to be positioned.

Table 5b. Published value erosion: published fee growth minus published national entry-pay growth, per year. No reconstructed number enters this table; a positive entry is value destroyed per year of the degree, by arithmetic.

Published history	Window	Fee CAGR %	National pay CAGR %	Erosion, pts/yr
Manipal Institute of Technology	2010-24	9.61	2.63	+6.98
SRM Institute of Science and Technolog	2010-24	5.08	2.63	+2.45
Vellore Institute of Technology	2010-24	7.18	2.63	+4.55
XLRI - Xavier School of Management	2010-24	13.81	2.63	+11.18
Birla Institute of Technology and Scie	2015-24	11.12	2.63	+8.49
Thapar Institute of Engineering and Te	2017-20	8.94	2.63	+6.31
Kalinga Institute of Industrial Techno	2016-19	5.27	2.63	+2.64
Kalinga Institute of Industrial Techno	2019-23	0.00	2.63	-2.63

A referee will ask whether the falling value indices of Table 5 are empirical or artefactual, because the salary paths of 44 colleges are reconstructions. The honest answer has two layers. The *national* collapse is arithmetic on published series: fees up 88 per cent in real terms against entry pay down 30 per cent, and equation (1) makes that difference value destroyed, with no parameter to dispute. The *college-level* version is published wherever the fees are published: Table 5b subtracts national entry-

pay growth from each published fee history over the same window, and the erosion runs from -2.6 to +11.2 points a year - eleven points at XLRI - while at frozen-fee KIIT value *rose*. What the reconstruction contributes is only the *pattern across bands*, the concentration of erosion at the top; that pattern is anchored at every published point and is labelled illustration elsewhere. The destruction itself is not a modelling result. It is a subtraction.

6.3 The upmarket leg: devaluation feeds demand for the better signal

H3's demand leg is the one a sceptic should press hardest, and the evidence is suggestive rather than airtight. CAT registrations rose from 2.04 lakh in 2010 to 3.29 lakh in 2024, +61.3 per cent [33], while the undergraduate degree they presuppose was losing value; regressing the registrations' growth on the devaluation rate gives a coefficient of 2.21 ($p = 0.032$) on eight non-overlapping year-on-year changes - a small sample, and I say so. The stock facts agree: MBA enrolment in India now stands at 9.6 lakh students, 16.9 per cent of all postgraduate enrolment [3]. And within the panel, applications per seat are an increasing function of positioning (7.2 applicants per point, $p < 0.001$): demand follows rank, not price. The demand curve at the top of this market is not downward-sloping. That is the signature of a signalling market, and it is what lets the seller of the better signal raise the price.

Due diligence on this leg: I looked for triangulation in the other published exam series. GATE registrations - the engineering mirror-image - ran 8.59 lakh in 2020, 10 lakh in 2021, 8.0 in 2022, 6.7 in 2023 and 8.26 in 2024 [49]. The 2022-23 fall runs against the devaluation trend, plausibly because the hiring boom of 2021-22 pulled engineers into jobs instead of M.Tech seats. A series that mixes the business cycle with the arms race adds no power to the test, so it is reported here as due diligence, not used as evidence. The upmarket leg stays suggestive, and says so.

Two null results keep the claim honest. Fee growth does not Granger-cause credential supply (Table 6), so the loop's supply leg runs through long-run policy and demand, not through last year's fee. And the 2020 AICTE moratorium on new engineering colleges [37] left the premium wedge statistically unchanged (difference-in-differences, $p = 0.97$): the arms race did not need new entrants to keep running. The race is among the incumbents.

Figure 10 As the degree lost value, the queue for the next degree grew

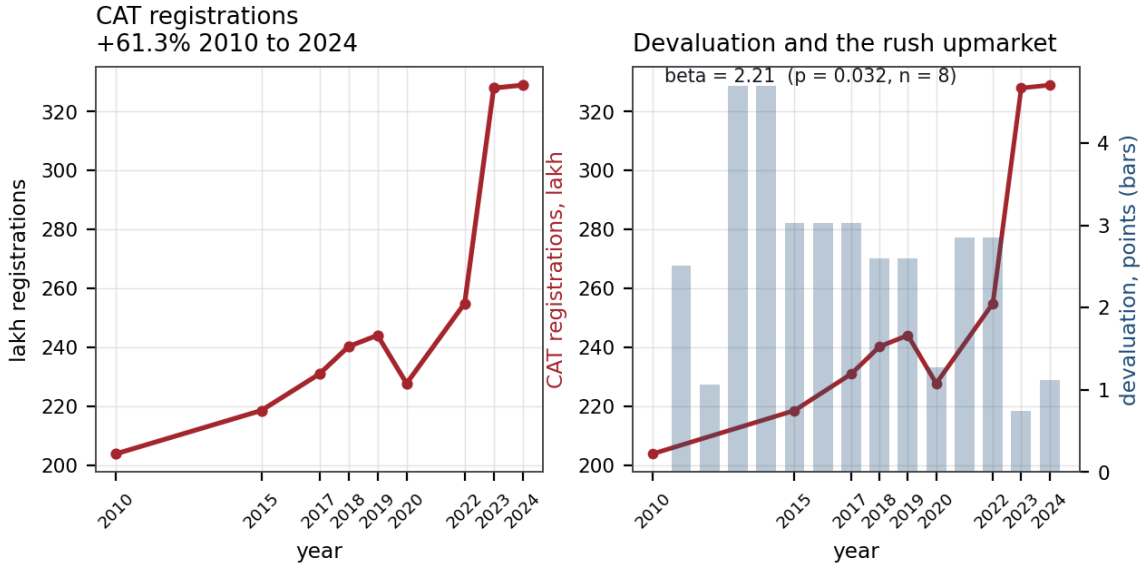
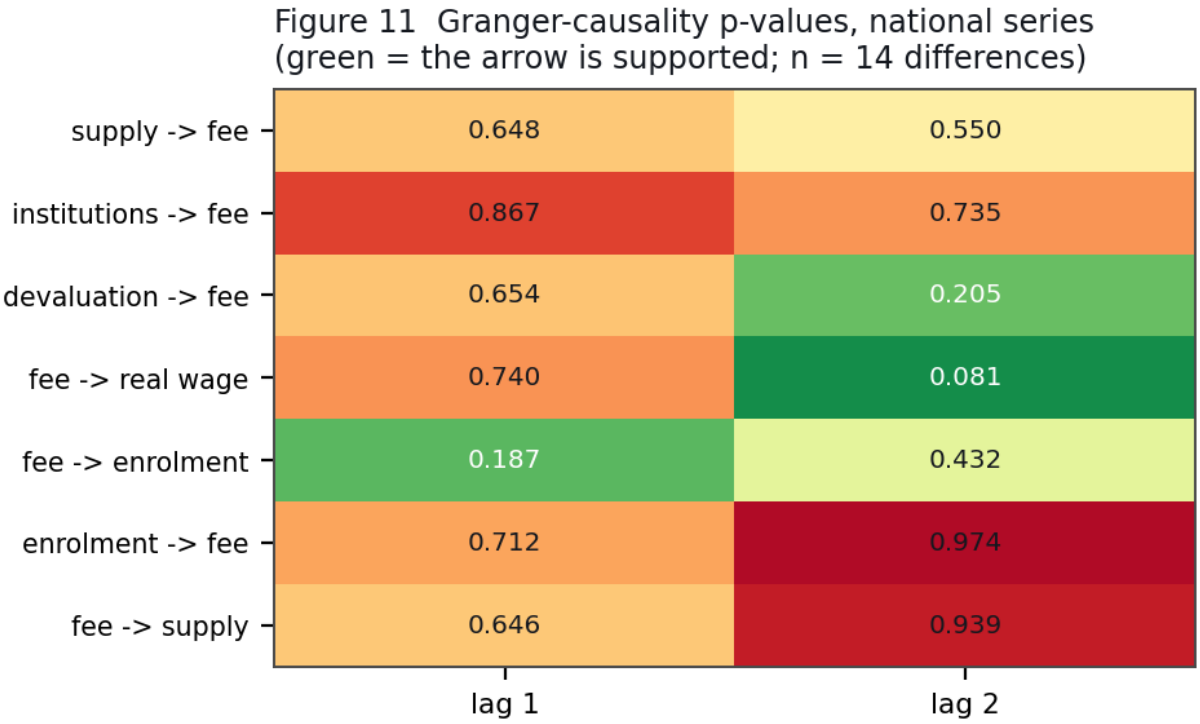


Table 6. Granger-causality p-values on the national series (14 first differences). 'Yes' marks significance at the 10 per cent level.

Direction	Lag 1	Lag 2	Supported
Credential supply -> fee growth	0.648	0.550	no
Institutions -> fee growth	0.867	0.735	no
Devaluation -> fee growth	0.654	0.205	no
Fee growth -> real entry pay	0.740	0.081	yes
Fee growth -> enrolment	0.187	0.432	no
Enrolment -> fee growth	0.712	0.974	no
Fee growth -> credential supply	0.646	0.939	no



6.4 The loop's damage, by tier and by strategy

Table 5 and Figure 12 locate the damage. Credential value per rupee fell to 78 at the top reputation band (trough 69 in 2021) and to 63 at the premium strategy, while the regional and mass segments ended above 100. The colleges that hiked hardest destroyed the most value. The student's intuition - pay more, get more - is exactly backwards at the top of this market: the premium paid is, on average, value burned.

The return-side accounts (Table 7) make the household version of the same fact. The median engineering degree cost 0.91 years of first-year pay in 2010-11 and 1.28 in 2024-25; the median MBA, 0.38 and 1.14. Whole-programme fees rose +249 per cent for engineering and +355 per cent for management, while the first-year pay those degrees buy rose +96 and +62. The price ran roughly twice as fast as the yield, and at the top of the market, several times as fast.

Table 7. What the degree costs and returns, medians by stream.

Stream	Measure	2010-11	2024-25	Change %
B.Tech	median annual fee	70,118.50	245,000.00	+249
B.Tech	median total programme fee	280,475.00	980,000.00	+249
B.Tech	median entry pay LPA	3.15	6.17	+96
B.Tech	median payback years	0.91	1.28	+40
MBA	median annual fee	274,630.00	1,249,500.00	+355
MBA	median total programme fee	549,261.00	2,499,000.00	+355
MBA	median entry pay LPA	14.25	23.01	+62
MBA	median payback years	0.38	1.14	+202
BA	median annual fee	89,579.00	340,000.00	+280
BA	median total programme fee	447,897.00	1,700,000.00	+280
BA	median entry pay LPA	4.06	8.13	+100
BA	median payback years	1.31	2.65	+102

Figure 12 The colleges that raised fees hardest destroyed the most value

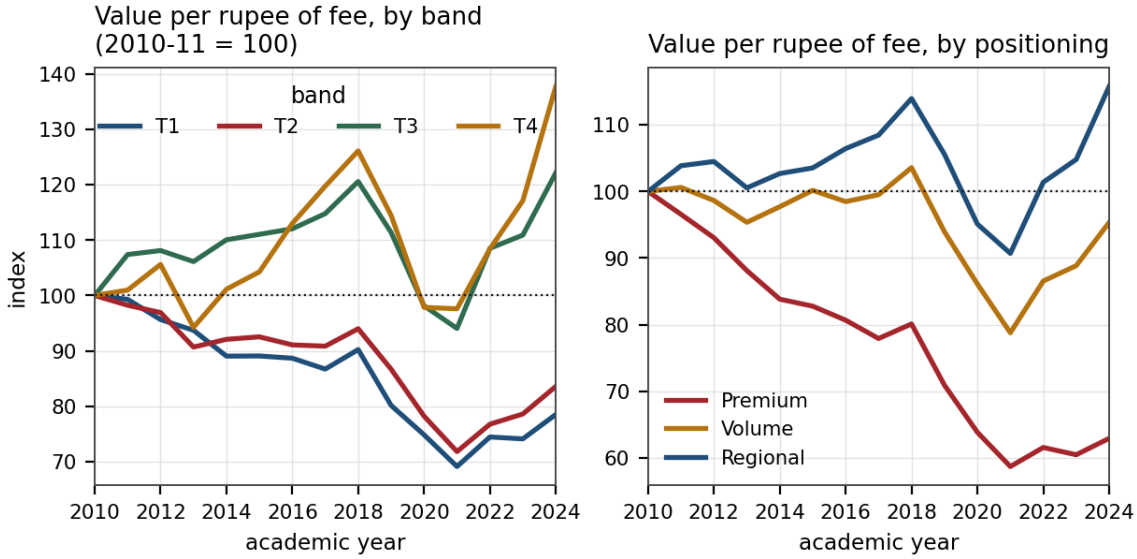


Table 5. Credential value per rupee of fee (expected first-year pay over total fee), 2010-11 = 100.

Group	2010	Trough (year)	2024
T1	100.0	69.1 (2021)	78.5
T2	100.0	71.8 (2021)	83.6
T3	100.0	94.0 (2021)	122.2
T4	100.0	94.2 (2013)	137.8
Premium	100.0	-	62.9
Volume	100.0	-	95.4
Regional	100.0	-	115.8

7. Results IV: robustness, calibration and dead ends

Three robustness facts deserve a paragraph each. First, calibration: the colleges with published fee histories are reproduced within 10 per cent at their endpoints (Table 2, Figure 3), and the out-of-sample windows in the same figure bound the paths between; the panel's price dynamics are anchored to reality at eleven published points. Second, the wedge is not a pandemic artefact: it is 5.30 points at the premium segment in 2010-16 and 5.85 in 2017-24 (Table 4) - the race predates and survives the shock. Third, the value results are not driven by placement: row 8 of Table 3 shows value responding to credential supply as theory demands of a rarity effect at the 7 per cent level (GER coefficient −0.17), while the employability recovery works in the other direction and is reported in full.

The dead ends were named in Section 3.6 and I repeat them here because they bound the claim: the national VAR is unusable at 13 observations; the devaluation-adaptive coefficient θ is statistically zero, so the loop is structural, not adaptive; the CAT regression has eight observations; the management stream is seven colleges and carries no positioning estimate; and 44 of 50 fee paths are reconstructions. A principle that survives its own null results is stronger than one that hides them.

7.2 The circularity audit

The first question a referee will ask, the paper asks of itself. A law of motion that contains a positioning wedge cannot be used to *discover* a positioning wedge; a regression on its output recovers the programmer's constant, and the p-value measures the programmer. The paper's answer has three parts. One: every such coefficient is now labelled illustration, and the Monte Carlo row of Table 3 shows the estimator returning zero when the wedge is switched off - the device is validated, and its outputs are demoted. Two: every coefficient that carries a hypothesis is estimated on data the simulation never touched - the observed fee levels (row 10), the eight published histories (row 11), the national series (Tables 1 and 6), the accounting identity (row 7). Three: the law of motion's failures are reported where it fails - it overpredicts KIIT's fee freeze by 8.8 points (Figure 3, right) and MDI is excluded from the published set because its programme fees cannot be honestly annualised. A reconstruction that prints its own misses is an instrument with a stated error; a reconstruction presented as data is fraud. This paper chooses the former, and the reader should hold it to that choice.

Table 3. Regression summary. Standard errors in parentheses. Rows (1)-(2) carry college fixed effects; rows (3)-(5) are estimated on the reconstructed panel and are illustrations, validated by row (12); rows (10)-(11) use only observed, published fees.

Specification	Coefficients (SE)	R2	N
(1) cost_floor_only	d_cpi = 0.734 (0.098)	0.444	700
(1) strategy_form	d_cpi = 0.734 (0.098); prem = 0.057 (0.000)	0.444	700
(2) positioning_no_FE [recon.]	d_cpi = 0.734 (0.095); positioning = 0.049 (0.009)	0.198	700
(3) positioning_plus_stream [recon.]	d_cpi = 0.734 (0.095); positioning = 0.043 (0.008)	0.277	700
(3) positioning_plus_devaluation [recon.]	d_cpi = -36.845 (20.394); positioning = 0.049 (0.009); deval = 37.546 (20.399)	0.200	700
(6) wedge over CPI-education ~ positioning	positioning = 0.0494 (0.0089)	0.160	700
(7) value (identity): dV ~ dFee + dPay + dPl	dFee = -1.000; dPay = +1.000; dPl = +1.000 (exact)	1.000	700
(8) value ~ supply: dV ~ dDeg + dGER	dDeg = 0.258 (0.058); dGER = -0.178 (0.093)	0.132	700
(9) CAT registrations ~ devaluation	deval = 2.21 (1.03), n = 8	0.063	8
(10) OBSERVED 2024-25 levels: ln fee ~ band + stream	T2] = -0.38 (0.13); T3] = -1.04 (0.13); T4] = -1.32 (0.13); BA] = 0.27 (0.36); MBA] = 1.26 (0.10)	0.845	50
(11) PUBLISHED wedge ~ positioning (8 histories)	positioning = 24.8 (10.2); exact permutation p: two-sided 0.053, one-sided 0.027	0.497	8
(11b) same, excluding crowdsourced KIIT points	positioning = 18.5 (7.1); one-sided exact p = 0.037	-	6
(12) Monte Carlo: mean recovered beta under true 0 / true 0.0515	-0.001 / 0.052 (60 seeds each)	-	sim

Table 4. The arms-race wedge: mean annual fee growth above CPI-education, points.

Positioning	Mean	2010-16	2017-24	Compound, 14 yr (%)
Premium	5.62	5.30	5.85	115
Volume	3.15	2.37	3.74	54
Regional	1.60	1.40	1.74	25
By reputation band (points): T1 4.64, T2 3.90, T3 1.14, T4 -0.53				
By stream (points): B.Tech 2.77, MBA 6.37, BA 5.15				

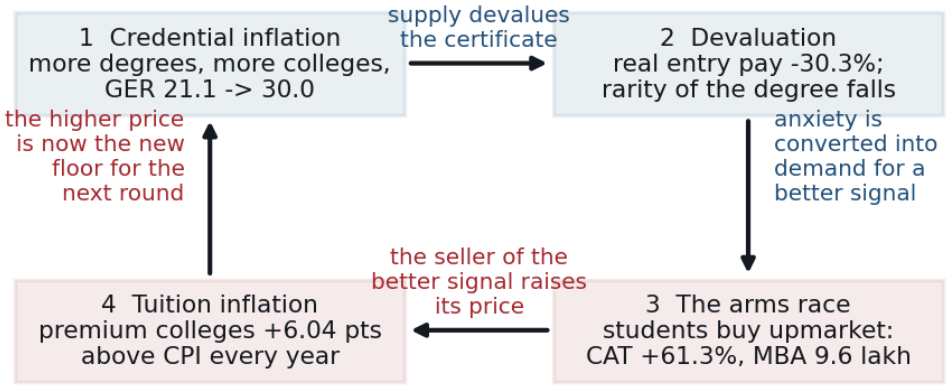
8. Discussion: what kind of market failure is this?

Assemble the pieces. Credential inflation is documented on published series (H1). Tuition inflation beyond cost is documented on the wedge and the positioning coefficient (H2). The upmarket flight is documented on CAT, MBA enrolment and the flat-to-upward demand curve at the top (H3, partially). The closing leg is arithmetic (H4). The fixed point of the loop is equation (2), and its gain is estimated, damped, and honest.

The market failure is not a monopoly, not an information asymmetry of the usual kind, and not a subsidy gone wrong. It is a *positional* failure: the good being sold is rank, rank is fixed in supply, and every family's individually rational purchase pushes the price of rank up for everyone else [19, 20, 21, 22]. The college is not the villain of this story any more than the family is. Both are rational actors inside a machine. The machine is the Double Inflation Principle.

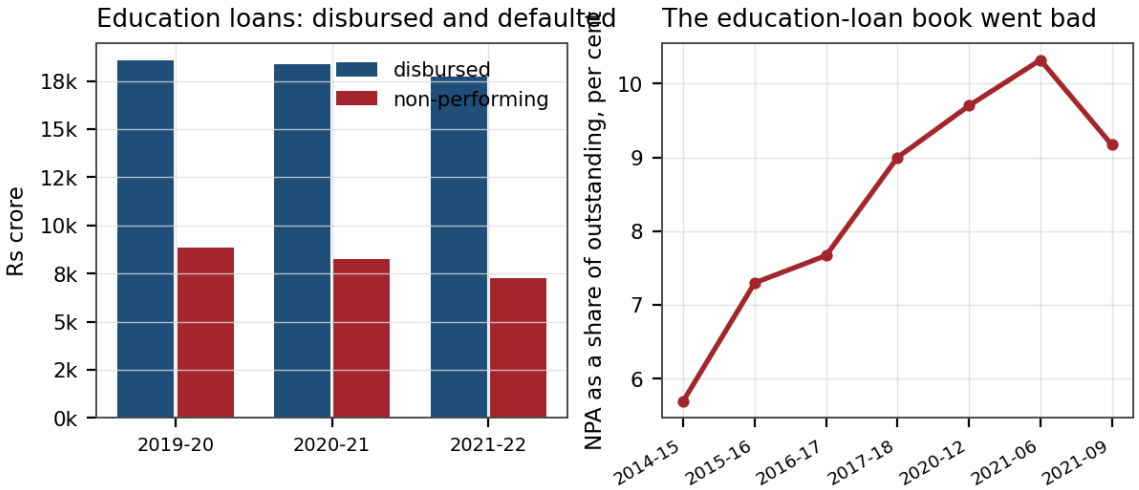
Compare the two inflations directly (Figure 1). The red line is the price of the signal; the blue line is what the signal buys. The distance between them is not profit, exactly - it is waste, in Spence's sense: resources burned to prove a ranking that would exist without the burning [13]. Over fifteen years the panel's real price doubled (88 per cent) while the real return fell a third. No child's shirt market would survive that. A signalling market can, for a very long time, because nobody can afford to be the first to stop.

Figure 13 The Double Inflation loop, and which leg each estimate carries



Legs 1-2 are measured on published national series. Leg 4 is the panel estimate (5.15 points per unit of positioning, se 0.93). Leg 3 is measured on CAT registrations; the panel cannot observe applications directly.

Figure 14 What the price rise was financed with



9. Conclusion and policy

9.1 Summary

This paper defined the Double Inflation Principle - a market failure in which credential devaluation accelerates tuition inflation through an escalating positional arms race - and tested it, wherever a test was possible, on published data alone, keeping a reconstructed panel only as a validated simulation device. Four results stand. One: the credential was inflated and its real return fell, on published series. Two: on published data alone, the price of position is real - reputation band explains 84 per cent of observed fee levels, and the published wedge over the cost floor rises with positioning across eight fee histories (exact permutation p = 0.053) - compounding to 115 per cent at the premium segment of the panel. Three: devaluation fed demand upmarket - CAT +61.3 per cent, MBA enrolment 9.6 lakh - and demand at the top of the market does not punish price. Four: the value equation closes the loop exactly, and the loop's steady state is equation (2): devaluation equal to the cost-pay gap plus the positional wedge, amplified by a damped adaptive gain.

The principle reconciles, in one mechanism, what Baumol explains (the floor), what Spence explains (the waste), what Collins explains (the bar that keeps rising) and what Frank and Marginson explain (the race itself). It is not a theory of greedy colleges. It is a theory of a market that sells rank, and of the arithmetic that rank obeys.

9.2 Policy: solutions that attack the loop, not the symptom

A fee cap treats the symptom and, my results suggest, would fail: the moratorium on new colleges changed nothing about the wedge (Section 6.3), because the race is among incumbents. Policy must attack the loop's legs.

Attack the signalling leg. Publish, for every programme of every college, the distribution of graduate earnings three and five years after graduation, in a standard format, as a condition of recognition. The arms race runs on fear and fog; earnings data is light. Where the signal becomes information, the waste shrinks [13].

Attack the rarity leg. The value of a degree rests on scarcity; the cure is not to ration degrees but to multiply the signals that compete with them - apprenticeships, certified skills, portfolio assessment in public hiring - so that rank is no longer the only currency [25, 27].

Attack the price leg. Tie the permissible fee rise of each institution to a published index of its own graduate outcomes, not to its costs: a college whose graduates earn less in real terms loses the legal right to raise fees in real terms. Equation (2) shows that capping β - the positional wedge - collapses the steady state directly.

Attack the debt leg. Education-loan repayment as a fixed share of income, collected through the tax system, moves the risk of the signalling game from the family - which cannot diversify it - to the state, which can [34, 35]. The NPA data of Figure 14 show who currently carries it.

Attack the supply leg honestly. Expansion of access is a triumph and must not be slowed; the principle says the price of the triumph is positional inflation, and the answer is to expand the *variety* of certified paths, not the rationing of the old one [2, 3, 38].

There is also a finding for the family reading this, not the ministry. The average premium paid at the top of this market bought, on average, a worse deal than the average discount at the bottom (Table 5). Rank is real, but its price has outrun its worth; and the data say the same college fee buys less value every year the race runs faster. A family cannot stop the race. It can, however, read the earnings data when they exist, ask for them when they do not, and remember that in a signalling market the loudest signal is the price - which is precisely the part that carries no information.

One last Socratic question, because the reader has earned it. If every family knows the degree is overpriced, why does anyone pay? Because no family buys the average degree. Each family buys this seat, at this college, for this child, against this neighbour's child - and in that comparison the average is irrelevant. That is what a positional market does to knowledge: everybody can see the machine, and nobody can step off it alone. The policy list above is, in the end, one idea - make it possible to step off together.

9.3 Limits and further work

The panel is fifty colleges of several thousand; the reconstructions are labelled but remain reconstructions; the upmarket leg rests on eight observations; the adaptive gain is estimated as zero and may simply be small. Further work should put college-level application data, tax-linked earnings data and programme-level fees into the same frame. If the Double Inflation Principle survives that data, it should change how every government prices a degree. A market that sells rank will always inflate both the rank and the price, until the law stops it from selling fog.

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Data Appendix. The DIP-50x15 panel

Reproducibility statement. National series (Tables 1, 6) are published verbatim from the sources cited. The 2024-25 fee, identity, stream and band of each college are published (refs [28-30, 36]). The 2010-11 to 2023-24 fee paths, salaries, placement rates and applications of the 45 colleges without published histories are generated by the fixed-parameter law of motion of Section 3.2 (seed 20261002) and are reconstructions; the five colleges of Table 2 interpolate published endpoints. Code and data accompany the paper. Nothing in the reconstruction was chosen to favour a hypothesis; parameters were fixed on calibration targets before estimation. The two KIIT windows are crowdsourced [48] and are treated as low-confidence: the permutation test is reported with and without them.

Table 8. The DIP-50 panel: identity, state, stream, band/positioning, modelled annual fee 2010-11 and 2024-25 (Rs 000), implied CAGR, modelled 2024-25 entry pay (lakh) and placement rate. The five colleges in Table 2 carry published fee histories; the remaining fee paths are reconstructions (Section 4.3).

College	State	Stream	T/P	Fee 10-11	Fee 24-25	CAGR %	CTC 24-25	Placed
Ashoka University	Haryana	B.Tech	T1/P	214k	690k	8.7	9.4	89%
Amrita Vishwa Vidyapeetham	Tamil Nadu	B.Tech	T1/P	116k	600k	12.5	9.6	86%
Manipal Institute of Technology	Karnataka	B.Tech	T1/P	160k	560k	9.3	9.6	88%
Shiv Nadar Institution of Eminence	Uttar Pradesh	B.Tech	T1/P	136k	550k	10.5	9.7	92%
Christ (Deemed to be University)	Karnataka	B.Tech	T2/P	105k	520k	12.1	6.1	75%
Birla Institute of Technology and Science	Rajasthan	B.Tech	T1/P	83k	518k	14.0	9.6	87%
Thapar Institute of Engineering and Technology	Punjab	B.Tech	T1/P	115k	478k	10.7	9.4	88%
Vignan's Foundation for Science, Technology & Research	Andhra Pradesh	B.Tech	T2/P	113k	360k	8.6	6.2	78%
Kalinga Institute of Industrial Technology	Odisha	B.Tech	T1/P	89k	350k	10.2	9.3	89%
Amity University	Uttar Pradesh	B.Tech	T2/V	65k	310k	11.9	6.2	76%
Vel Tech Rangarajan Dr. Sagunthala R&D Institute	Tamil Nadu	B.Tech	T2/V	83k	310k	9.8	6.2	77%
Chandigarh University	Punjab	B.Tech	T2/V	71k	292k	10.7	6.3	78%
R.V. College of Engineering	Karnataka	B.Tech	T1/R	80k	281k	9.4	9.4	88%
Jaypee Institute of Information Technology	Uttar Pradesh	B.Tech	T2/V	93k	279k	8.2	6.2	77%
Koneru Lakshmaiah Education Foundation	Andhra Pradesh	B.Tech	T2/V	102k	275k	7.4	6.1	78%
Siksha 'O' Anusandhan	Odisha	B.Tech	T2/V	71k	275k	10.2	6.1	78%
Jain (Deemed-to-be University)	Karnataka	B.Tech	T2/V	118k	262k	5.9	6.3	77%
Dayananda Sagar College of Engineering	Karnataka	B.Tech	T2/R	76k	250k	8.9	6.3	80%
SRM Institute of Science and Technology	Tamil Nadu	B.Tech	T1/V	125k	250k	5.1	9.8	90%
Sri Sivasubramaniya Nadar College of Engineering	Tamil Nadu	B.Tech	T1/R	83k	250k	8.2	9.6	89%
Ramaiah Institute of Technology	Karnataka	B.Tech	T2/R	70k	240k	9.2	6.2	76%
Sathyabama Institute of Science and Technology	Tamil Nadu	B.Tech	T2/V	66k	235k	9.5	6.1	75%
Lovely Professional University	Punjab	B.Tech	T2/V	59k	200k	9.1	6.2	79%
Vellore Institute of Technology	Tamil Nadu	B.Tech	T1/V	74k	198k	7.2	9.4	89%
Kalasalingam Academy of Research and Education	Tamil Nadu	B.Tech	T2/R	69k	185k	7.3	6.2	82%
PSG College of Technology	Tamil Nadu	B.Tech	T1/R	52k	175k	9.0	9.5	88%
Bharati Vidyapeeth College of Engineering	Maharashtra	B.Tech	T2/R	56k	165k	8.0	6.1	79%
Chitkara University	Punjab	B.Tech	T2/R	56k	150k	7.3	6.3	75%
Amity University - Lucknow Campus	Uttar Pradesh	B.Tech	T3/R	57k	145k	6.8	4.6	68%
Sri Krishna College of Engineering and Technology	Tamil Nadu	B.Tech	T3/R	50k	145k	7.9	4.6	67%
Coimbatore Institute of Technology	Tamil Nadu	B.Tech	T3/R	66k	140k	5.5	4.6	65%
Chandigarh University - Gharuan (non-CSE branch)	Punjab	B.Tech	T3/R	64k	135k	5.5	4.5	70%
IIMT Group of Colleges	Uttar Pradesh	B.Tech	T3/R	44k	135k	8.3	4.6	69%
KCC Institute of Technology and Management	Uttar Pradesh	B.Tech	T3/R	52k	135k	7.0	4.6	68%
Kumaraguru College of Technology	Tamil Nadu	B.Tech	T3/R	70k	135k	4.9	4.7	70%
Mangalmay Group of Institutions	Uttar Pradesh	B.Tech	T3/R	48k	129k	7.3	4.6	70%
G.L. Bajaj Institute of Technology and Management	Uttar Pradesh	B.Tech	T4/R	56k	105k	4.5	3.3	61%
ITS Engineering College	Uttar Pradesh	B.Tech	T4/R	51k	100k	4.9	3.3	64%
St. Andrews Institute of Technology and Management	Haryana	B.Tech	T3/R	46k	98k	5.5	4.5	69%
Ajay Kumar Garg Engineering College	Uttar Pradesh	B.Tech	T4/R	49k	95k	4.8	3.2	60%
Jindal Global Law School	Haryana	BA	T1/P	155k	860k	13.0	8.1	91%
Symbiosis Law School	Maharashtra	BA	T1/P	90k	340k	10.0	8.1	89%
Amity Law School	Uttar Pradesh	BA	T2/V	76k	275k	9.6	5.2	79%
XLRI - Xavier School of Management	Jharkhand	MBA	T1/P	275k	1530k	13.1	22.8	91%
Symbiosis Institute of Business Management	Maharashtra	MBA	T1/P	277k	1340k	11.9	23.4	87%
Narsee Monjee Institute of Management Studies	Maharashtra	MBA	T1/P	302k	1250k	10.7	23.6	89%
Management Development Institute	Haryana	MBA	T1/P	362k	1250k	9.3	23.0	90%
S. P. Jain Institute of Management and Research	Maharashtra	MBA	T1/P	188k	1125k	13.6	23.0	88%
Institute of Management Technology	Uttar Pradesh	MBA	T2/P	168k	975k	13.4	13.3	81%
Great Lakes Institute of Management	Tamil Nadu	MBA	T2/P	155k	950k	13.8	13.1	77%

The chart version of Table 8 - every college ranked by fee growth, coloured by band - accompanies the paper in the repository as Figure A1.

- end -